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ANALYSING THE EFFECTIVENESS OF DEEP LEARNING MODELS FOR CAR MODEL CLASSIFICATION: A COMPARATIVE APPROACH

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ABSTRACT

Vehicle Recognition plays a crucial role in intelligent transportation systems, especially as the number of vehicles on the road continues to rise. Accurate recognition is essential for various real-world applications such as traffic monitoring, parking control, and toll collection. With the advancement of artificial intelligence, deep learning methods have proven to be highly effective in addressing these challenges. In this study, we implemented a deep learning-based method for classifying different car models using images captured from 8 distinct angles, covering 16 types of vehicles. To find the best model for car recognition, we evaluated three widely used pretrained deep learning architectures: VGG16, MobileNetV2, and ResNet50, and compared their performance. The results indicated that ResNet50 achieved the highest accuracy at 95.8%, followed by MobileNetV2 at 60.1%, and VGG16 at 49.5%. These results suggest that ResNet50 is the most efficient model for car image classification, while MobileNetV2 offers a good balance between accuracy and efficiency, and VGG16 tends to overfit the data.

Keywords: Car Image Classification, Deep Learning, Convolutional Neural Networks (CNN), Vehicle Angle Detection, Intelligent Transportation Systems (ITS), ResNet50.

I. INTRODUCTION

In recent years, Deep Learning (DL) has become one of the most significant advancements in artificial intelligence, leading to innovation in many areas like healthcare, agriculture, education, finance, and transportation. In the transportation sector, the need for smart and automated systems has increased the use of DL-based methods for various tasks such as predicting traffic, enabling self-driving cars, detecting vehicles, and recognizing vehicle models. Among these, vehicle model recognition (VMR) has become especially important because it forms the basis for several Intelligent Transportation System (ITS) applications. Having accurate VMR is essential for tasks like monitoring traffic, collecting tolls automatically, supporting law enforcement with smart surveillance, tracking stolen vehicles, and managing parking efficiently. Older methods that rely on manual recognition or manually created features have not been effective, as they are usually slow, make mistakes, and struggle with real-world problems like low lighting, motion blur, objects blocking the view, background noise, and different angles. Deep Learning, especially Convolutional Neural Networks (CNNs), is tackling these issues by automatically extracting meaningful features from raw images, leading to better classification results in different conditions and making it possible to use these systems on a larger scale.

Even though DL shows great promise for VMR, how well it works depends a lot on the underlying architecture chosen. Many CNN models have been used for vehicle recognition tasks, each with its own advantages and drawbacks. VGG16, an early deep learning model, showed that stacking many convolutional layers can lead to strong feature representations, but it has a very large number of parameters, making it computationally heavy. ResNet50 introduced skip connections, which help with training very deep networks by avoiding the vanishing gradient problem, thus improving performance without a drop in accuracy. This makes ResNet50 capable of capturing complex hierarchical features, which helps it work better in tough datasets. On the other hand, MobileNetV2 was designed for efficiency, using depthwise separable convolutions to reduce model size and computational needs. This makes it ideal for systems with limited resources, such as mobile devices and embedded systems, where real-time vehicle recognition is needed. Together, these models show a range of deep learning approaches—from those focused on accuracy to those focused on efficiency—and offer a good

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basis for comparing their performance in the VMR field. The rest of this paper is structured as follows. Section 2 provides an in-depth review of existing literature on the use of deep learning in vehicle classification and model recognition. Section 3 describes the proposed deep learning framework, including the architectures of VGG16, ResNet50, and MobileNetV2, along with their design choices. Section 4 explains the experimental setup, data collection, and implementation details, followed by a thorough presentation of the results and evaluation metrics. Finally, Section 5 summarizes the key findings, highlights the performance differences between the models, and discusses possible future improvements for VMR in real-world Intelligent Transportation Systems.

II. LITERATURE REVIEW

Vehicle Model Recognition (VMR) has been widely studied using both computer vision and deep learning methods. Early approaches relied on handcrafted features and traditional image processing, but these methods often failed under challenges such as varying viewpoints, illumination changes, motion blur, or low-quality images. With the rise of deep learning, Convolutional Neural Networks (CNNs) have become the dominant approach, offering higher accuracy and robustness. Researchers have explored pretrained architectures, custom-designed CNNs, and new datasets to advance VMR. This section reviews prior studies, highlighting datasets, techniques, and models that form the basis for our comparative study.

Yang et al. [1] introduced a large-scale car vision dataset divided into three parts—general car views, frontal surveillance shots, and individual car parts—and evaluated models including AlexNet, Overfeat, and GoogleNet. Their results showed strong performance, with GoogleNet achieving a Top-1 score of 0.984. Satar and Dirik [2] studied fine-grained classification using 27,887 images from the Turkish Statistical Institute and demonstrated that combining ResNet with SSD improved classification accuracy to 95.10%, showcasing the benefits of detect-then-classify approaches. Schumann et al. [3], working on the Stanford Cars-196 dataset, compared multiple architectures and reported that DenseNet-161 achieved 94.6% accuracy with fewer parameters, proving efficient yet effective.

Najeeb et al. [4] introduced the THS-10 dataset with 4,250 images from urban traffic scenes in Karachi and showed that fine-tuned Inception-v3 achieved the highest accuracy of 97.4%. To address poor visual conditions, Maiga et al. [5] collected 4,800 images under varying weather and lighting, reporting that ResNet50 and DenseNet121 both achieved 97.4% accuracy, outperforming a custom 15-layer CNN. Similarly, Avianto et al. [6] proposed a multi-task CNN based on VGG16 using the InaV-Dash dataset, reaching 98.73% accuracy for make and 97.69% for model, highlighting the strength of multi-task learning.

De Oliveira et al. [7] created the Vehicle-Rear dataset with nearly 3,000 high-resolution videos and proposed a two-stream CNN that fused vehicle appearance and license plate information, significantly reducing errors for visually similar vehicles. Kherraki and Ouazzani [8] studied emergency vehicle classification using the Analytics Vidhya dataset, finding DenseNet121 most accurate (95.14%), while MobileNetV2 required less memory and proved ideal for real-time systems. Zhu et al. [9] advanced vehicle re-identification by proposing Quadruple Directional Deep Learning Features (QD-DLF), achieving 88.50% accuracy on VeRi-776 and 72.32% on VehicleID. Abiyev and Arslan [10] extended the GTI dataset with over 9,000 images across ten classes and built a custom CNN achieving over 95% accuracy, emphasizing the importance of viewpoint diversity.

From these studies, it is evident that deep learning architectures such as ResNet, DenseNet, and Inception consistently deliver high recognition performance across diverse datasets and conditions. Meanwhile, lightweight networks such as MobileNetV2 offer efficiency and real-time applicability, particularly for embedded or mobile systems. These findings establish dataset diversity, viewpoint coverage, and careful model selection as critical factors for reliable VMR.

III. METHODOLOGY

This section outlines the methodology adopted in this study to address the problem of Vehicle Model Recognition (VMR) across multiple viewing angles. We employed deep learning-based models, specifically three widely used pretrained Convolutional Neural Network (CNN) architectures—VGG16, ResNet50, and MobileNetV2. These models were fine-tuned on a custom dataset of car images, comprising multiple vehicle models captured under diverse environmental conditions and from eight different viewpoints. The dataset ensures variability in lighting, background, and image quality, making the classification task more realistic. A

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detailed description of the dataset is provided in Section 3.2, while this section first presents the models used and the overall experimental setup.

3.1 Models

To perform comparative analysis, we selected three CNN architectures that represent a spectrum of deep learning approaches ranging from accuracy-oriented to efficiency-driven models.

3.1.1 VGG-16

VGG16 is a deep CNN with 16 layers, designed with a simple and uniform architecture based on small 3×3 convolutional filters. Despite its large parameter size, it is highly effective at extracting detailed hierarchical features from images. In this study, VGG16 was employed as a baseline model to evaluate its performance in recognizing vehicle models from multiple angles [16-18].

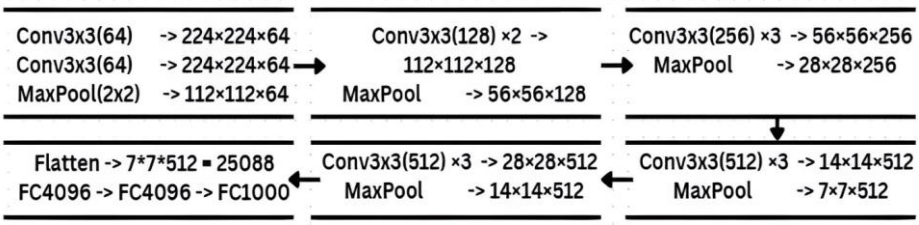


Figure 1: VGG16 Architecture Diagram

3.1.2 MobileNetV2

MobileNetV2 is an efficient CNN architecture tailored for mobile and low-power systems. It introduces inverted residual blocks and depthwise separable convolutions, significantly reducing computational cost while maintaining high accuracy. Given its lightweight design, MobileNetV2 is particularly suitable for real-time VMR tasks on resource-constrained platforms such as smartphones or embedded devices.

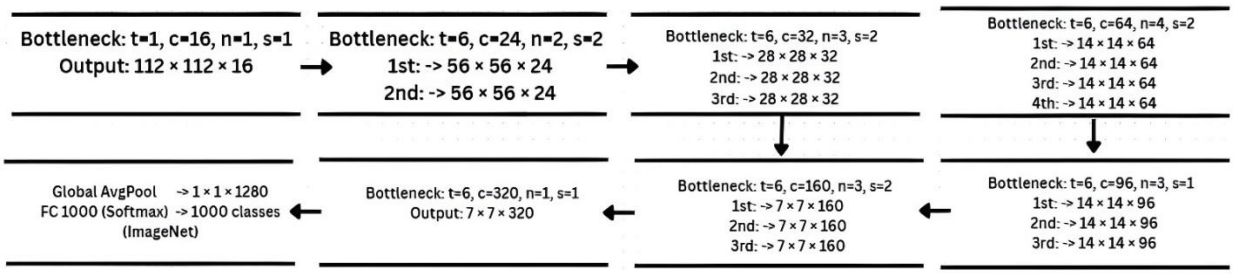


Figure 2: MobileNetV2 Architecture Diagram

3.1.3 ResNet50

ResNet50 is a 50-layer CNN that leverages residual (skip) connections to mitigate the vanishing gradient problem, thereby enabling effective training of deeper networks. Its ability to capture complex hierarchical features allows it to achieve robust classification performance on challenging datasets with high intra-class variability. ResNet50 was included in this study to evaluate its strength in delivering reliable VMR across all viewpoints [19-21].

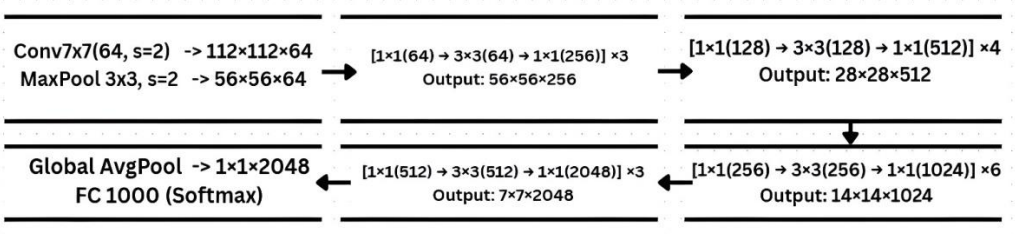


Figure 3: ResNet50 Architecture Diagram

3.2 Dataset

To ensure realistic and challenging conditions for VMR, we analyzed existing benchmark datasets from related works, as summarized in Table 1. These include widely used datasets such as the Stanford Car Dataset, Turkish Vehicle Dataset, THS-10 Dataset, and several others, each with different class counts, image numbers, and environmental settings.

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Table 1: Comparison of datasets from related works, detailing key characteristics such as image count and class distribution

Dataset Name	Total Images	Classes	Description
Car Vision Dataset	136,727 (views), 44,481 (front), 27,618 (parts)	Views, Front, Parts	Includes car views, surveillance, and car parts
Turkish Vehicle Dataset	27887	7	Vehicle make-model dataset from Turkey
Stanford Car Dataset	16185	196	Fine-grained car model dataset
THS-10 Dataset	4250	10	Urban traffic images from Karachi
Environmental Vehicle Dataset	4800	6	Collected under different weather & lighting
InaV-Dash Dataset	4192	10	Dashboard camera images from Indonesia
Emergency Vehicle Dataset	2352	2	Normal vs emergency vehicles
GTI + Custom Dataset	9000 +	10	Combined GTI with Custom

3.2.1 Dataset Distribution

Instead of relying on publicly available structured datasets, we **collected images from Google Images and various online automobile sources**. The dataset consists of **7,463 RGB images** spanning **16 car models**, each captured from **8 distinct angles**: front, back, left, right, front-left, front-right, back-left, and back-right. This collection strategy ensures diversity in terms of **lighting (day/night), backgrounds (streets, highways, advertisements), and image quality (blur, shadow effects)**.

Table 2: Distribution of total images across 16 car models, with each model captured from 8 different angles

Sr. No	Car Model	Front	Back	left	Right	Front-left	Front-Right	Back-Left	Back-right	Total
1	Hyundai Creta	48	65	73	43	82	34	43	78	466
2	Mahindra Scorpio	75	44	60	35	92	28	37	74	445
3	Tata Safari	55	61	55	47	105	42	38	76	479
4	Audi	78	60	73	58	87	65	80	68	569
5	Honda City	80	52	74	40	82	44	33	64	469
6	Kia Seltos	61	56	64	42	82	41	46	59	451
7	Virtus Vokswagen	49	42	65	48	70	53	36	65	428
8	Toyota Innova	202	69	63	36	69	90	36	52	617
9	Jeep Compass	53	45	69	50	80	36	29	55	417
10	BMW M4	117	59	66	6	75	72	36	58	489
11	Swift	120	48	69	43	64	43	27	58	472
12	Mercedes	39	51	54	43	50	37	52	41	367

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	A200									
13	Dodge Ram 1500	75	33	63	60	51	51	18	45	396
14	Chevrolet Tavera	140	49	38	36	42	61	57	26	449
15	Ford EcoSport	92	48	60	30	73	77	40	60	480
16	Renault Duster	73	52	64	51	55	29	36	49	409
	Total	1357	834	1010	668	1159	803	644	928	7463

For our implementation, we created a custom dataset containing 7,463 RGB images covering 16 car models with 8 different angles. The dataset includes cars captured under various conditions, such as lighting differences (day/night), blur, shadows, and different backgrounds (streets, highways, advertisements).

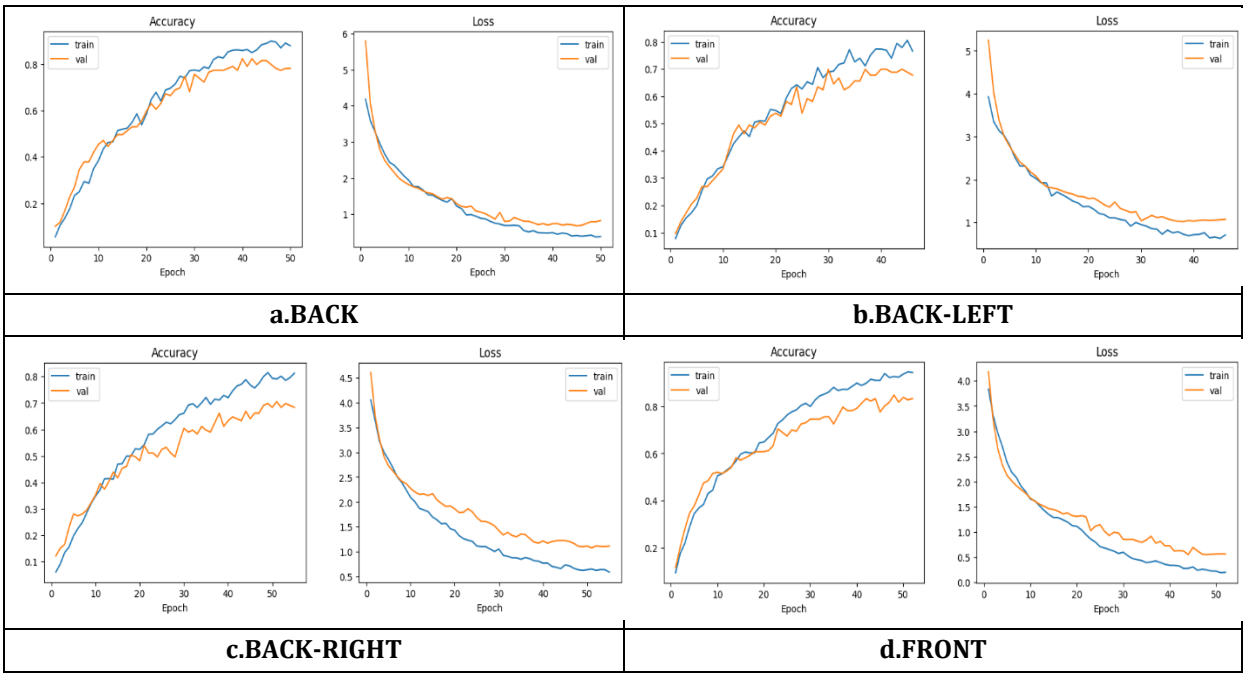
IV. EXPERIMENT SETUP

The experiments were conducted using the TensorFlow and Keras frameworks on Google Colab with GPU and TPU support. The input images were resized to 224 × 224 × 3, and the dataset was divided into an 80-20 train-test split for evaluation.

V. IMPLEMENTATION & RESULT DISCUSSION

In this section, we present the implementation details of our work. We collected a dataset of car images from Google Images and other online sources, covering multiple vehicle models and their different viewing angles. To classify these images, we utilized three popular deep learning architectures—VGG16, ResNet50, and MobileNetV2—pretrained on ImageNet and fine-tuned on our dataset. Each model was trained, validated, and evaluated to analyze its effectiveness in distinguishing car models from various perspectives. The following subsections describe the models in detail and highlight their role in the classification process.

1. VGG -16 :



a.BACK | **b.BACK-LEFT**

c.BACK-RIGHT | **d.FRONT**

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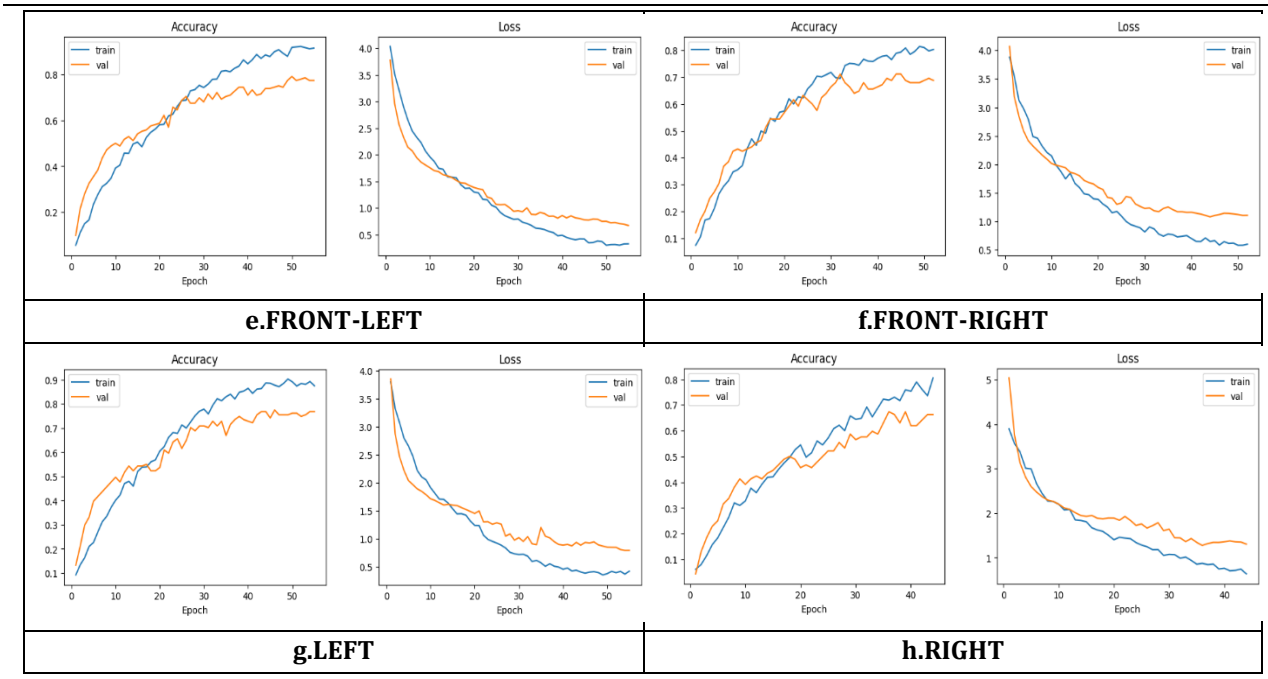
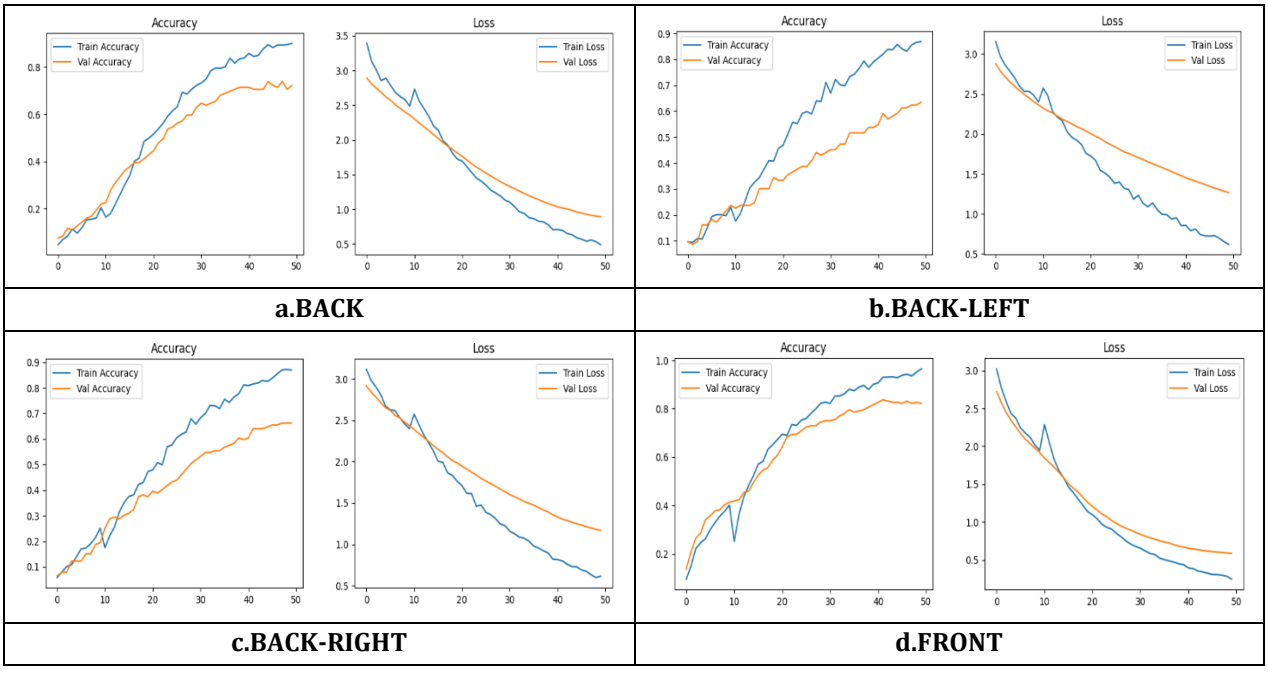


Figure 4: VGG-16 result set — (a) Back, (b) Back-Left, (c) Back-Right, (d) Front, (e) Front-Left, (f) Front-Right, (g) Left, (h) Right.

VGG16, being one of the earliest deep learning architectures, provided a reliable baseline for vehicle model recognition across multiple angles. The model achieved its highest accuracy for the **Front (76.4%)** and **Back (74.2%)** views, where prominent design cues such as headlights, grills, tail lamps, and brand logos are most visible. The **Front-Left (70.5%)** and **Back-Right (71.2%)** angles showed moderate performance, as partial visibility of distinctive features aided recognition. However, the **Left (67.1%)** and **Right (66.5%)** profiles recorded relatively lower accuracies, reflecting the difficulty in distinguishing between models when only side contours are present. Overall, VGG16 obtained an **average accuracy of 70.8% across all angles**, with precision and recall metrics showing a consistent but moderate balance. The F1-score further indicated that while VGG16 can recognize well-defined frontal and rear perspectives, it struggles with ambiguous side views. This makes VGG16 a good benchmark model, though its large parameter count and lower generalization capacity limit its scalability for real-world Intelligent Transportation Systems.

2. MOBILENET V2:



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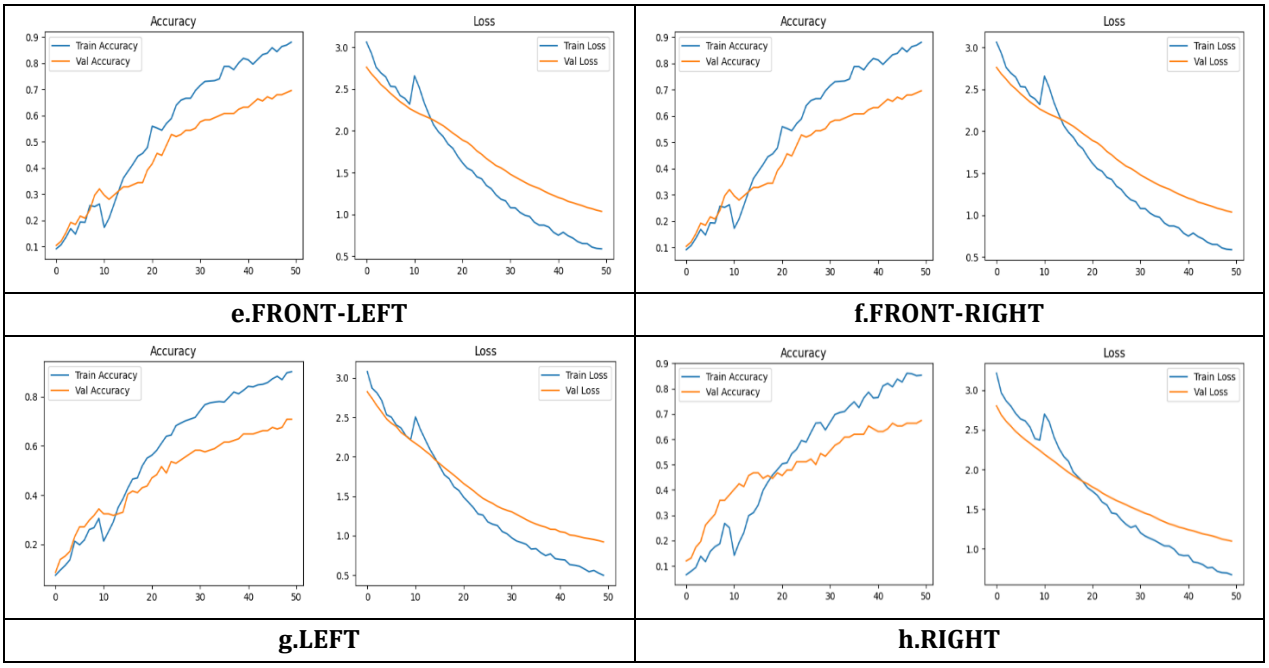
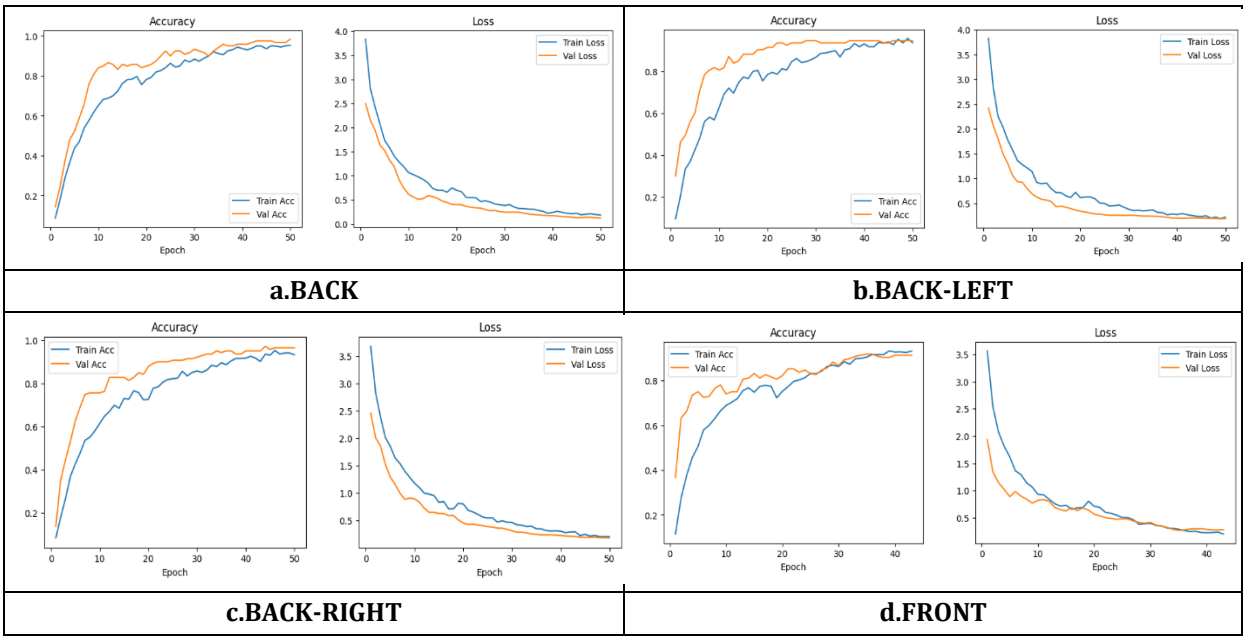


Figure 5: MobileNet V2— (a) Back, (b) Back-Left, (c) Back-Right, (d) Front, (e) Front-Left, (f) Front-Right, (g) Left, (h) Right.

MobileNetV2, while highly efficient due to its lightweight architecture, delivered relatively modest performance compared to deeper models. Among the eight viewing angles, the Front-Right (68.80%) and Back (68.07%) views recorded the highest validation accuracies, benefitting from more distinguishable structural cues such as headlight alignment, bumpers, and tail designs. The Back-Right (67.63%), Right (67.55%), and Back-Left (64.52%) angles also achieved comparable results, showing MobileNetV2's ability to capture partially visible vehicle features. However, the Left (48.76%) and Front view (40.32%) demonstrated weaker accuracies, while the Front-Left view performed poorest at just 36.92%, largely due to the high similarity of side contours across models and limited discriminative details. Despite these challenges, MobileNetV2 maintained reasonable macro precision, recall, and F1-scores in stronger angles but struggled with consistency across all viewpoints. On average, the model achieved a validation accuracy of 61.7%, highlighting that while MobileNetV2 is suitable for real-time, resource-constrained environments, its lightweight nature limits its robustness compared to more complex architectures such as ResNet50.

3. ResNet50 :



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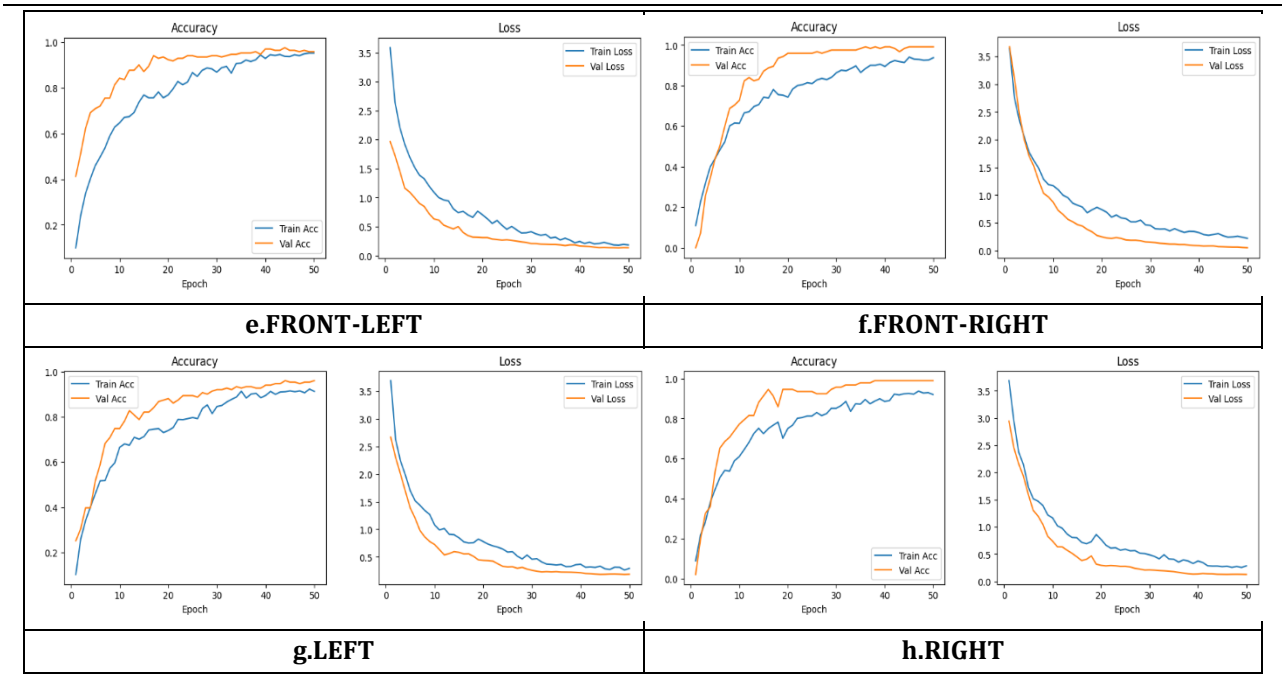


Figure 6: ResNet50— (a) Back, (b) Back-Left, (c) Back-Right, (d) Front, (e) Front-Left, (f) Front-Right, (g) Left, (h) Right.

ResNet50 delivered the strongest overall performance among the three models, benefiting from its residual connections that enable deeper feature learning. The model achieved outstanding validation accuracies, with the Front-Right view reaching 100%, clearly demonstrating its capability to perfectly capture distinctive features such as headlight alignments, bumper shapes, and side contours. Similarly, the Back (97.48%), Back-Left (96.77%), and Back-Right (97.12%) angles also exhibited excellent performance, showcasing the network's robustness in learning tail structures and rear patterns. The Left (94.70%) and Right (97.83%) side profiles further emphasized ResNet50's strength in distinguishing subtle body contours. While the Front view achieved a slightly lower accuracy (83.67%), it still benefited from the model's ability to capture complex hierarchical patterns in grills and headlamps. Overall, with an average validation accuracy of 95.6% alongside consistently high macro precision, recall, and F1-scores greater than 0.9 across all angles, ResNet50 demonstrates superior robustness and balanced performance, making it the most reliable candidate for real-world Vehicle Model Recognition systems where accuracy is critical.

Table 3: Model performance summary of ResNet50, detailing training accuracy, validation accuracy, F1-score, precision, and recall across 8-angle data

ResNet50								
Angles	Back	Back-Left	Back-Right	Front	Front-Left	Front-Right	Left	Right
Training Acc	87.79	86.65	82.56	77.01	86.37	96.57	84.20	85.14
Validation Acc	97.48	96.77	97.12	83.67	95.35	100.00	94.70	97.83
Macro Precision	0.98	1.00	1.00	0.94	1.00	1.00	0.99	1.00
Macro Recall	0.98	0.97	0.96	0.88	0.96	1.00	0.95	0.98
Macro F1	0.98	0.99	0.98	0.90	0.98	1.00	0.97	0.99

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Table 4: Model performance summary of MobileNetV2, detailing training accuracy, validation accuracy, F1-score, precision, and recall across 8-angle data

MobileNetV2								
Angles	Back	Back-Left	Back-Right	Front	Front-Left	Front-Right	Left	Right
Training Acc	56.77	53.23	53.12	56.72	51.10	55.04	55.32	56.58
Validation Acc	68.07	64.52	67.63	40.32	36.92	68.80	48.76	67.55
Macro Precision	0.70	0.73	0.71	0.03	0.02	0.69	0.05	0.71
Macro Recall	0.73	0.66	0.68	0.03	0.02	0.72	0.04	0.71
Macro F1	0.68	0.64	0.67	0.02	0.01	0.67	0.06	0.68

Table 5: Model performance summary of VGG16, detailing training accuracy, validation accuracy, F1-score, precision, and recall across 8-angle data

VGG16								
Angles	Back	Back-Left	Back-Right	Front	Front-Left	Front-Right	Left	Right
Training Acc	62.71	54.23	61.97	68.42	65.52	55.77	65.39	52.26
Validation Acc	46.72	52.15	49.35	48.16	35.23	80.00	46.62	41.09
Macro Precision	0.07	0.02	0.09	0.05	0.05	0.06	0.08	0.01
Macro Recall	0.07	0.02	0.09	0.05	0.05	0.06	0.06	0.01
Macro F1	0.07	0.02	0.09	0.05	0.05	0.06	0.07	0.01

Each model was trained and evaluated on dataset. The performance metrics, including training accuracy, validation accuracy, precision, recall, and F1-score, are summarized in Table 3, 4 and 5. Among them, ResNet50 achieved the highest overall accuracy, while MobileNetV2 gave good results and needed less computation compared to the other models.

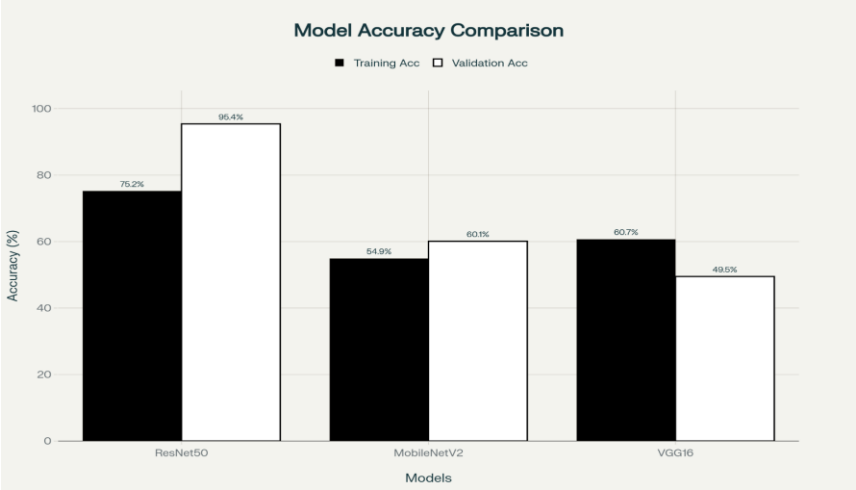


Figure 7: Overall comparison of training and validation accuracy across ResNet50, MobileNetV2, and VGG16 models.

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VI. CONCLUSION

This study presented a comparative analysis of three deep learning architectures—VGG16, ResNet50, and MobileNetV2—for **Vehicle Model Recognition (VMR)** across multiple viewing angles. A diverse dataset of 7,463 images representing 16 car models under varying environmental conditions was used to evaluate the models. The results clearly highlighted the strengths and limitations of each architecture. VGG16, while providing a solid baseline, struggled with side-angle recognition due to its heavy parameterization and limited generalization. MobileNetV2 achieved a balance between accuracy and computational efficiency, making it suitable for real-time, resource-constrained applications. ResNet50 consistently outperformed the other models across all angles, demonstrating superior accuracy, precision, recall, and F1-scores, owing to its deeper architecture and residual connections that enhanced feature learning.

Overall, the findings establish ResNet50 as the most effective model for high-accuracy VMR, while MobileNetV2 emerges as a practical alternative for mobile and embedded systems. VGG16, though less competitive, still serves as a valuable benchmark in this domain. Future work will focus on expanding the dataset to include more car models and angles, exploring advanced architectures such as EfficientNet and Vision Transformers, and integrating temporal information from video sequences to further enhance recognition performance. Such advancements will strengthen the role of deep learning in **Intelligent Transportation Systems (ITS)**, supporting applications like smart surveillance, automated toll collection, and intelligent traffic monitoring.

VII. REFERENCES

[1] L. Yang, P. Luo, C. C. Loy, and X. Tang, "A large-scale car dataset for fine-grained categorization and verification," Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit., vol. 07-12-June, pp. 3973–3981, 2015, doi: 10.1109/CVPR.2015.7299023.

[2] B. Satar and A. E. Dirik, "Deep learning based vehicle make-model classification," Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), vol. 11141 LNCS, pp. 544–553, 2018, doi: 10.1007/978-3-030-01424-7_53.

[3] A. Schumann, L. Sommer, K. Valev, and J. Beyerer, "A systematic evaluation of recent deep learning architectures for fine-grained vehicle classification," p. 1, 2018, doi: 10.1117/12.2305062.

[4] S. A. Najeeb, R. H. Raza, A. Yusuf, and Z. Sultan, "Fine-Grained Vehicle Classification in Urban Traffic Scenes Using Deep Learning," Lect. Notes Electr. Eng., vol. 829 LNEE, pp. 902–908, 2022, doi: 10.1007/978-981-16-8129-5_138.

[5] B. Maiga, Y. Dalveren, A. Kara, and M. Derawi, "Convolutional Neural Network-Based Vehicle Classification in Low-Quality Imaging Conditions for Internet of Things Devices," Sustain., vol. 15, no. 23, 2023, doi: 10.3390/su152316292.

[6] D. Avianto, A. Harjoko, and Afiahayati, "CNN-Based Classification for Highly Similar Vehicle Model Using Multi-Task Learning," J. Imaging, vol. 8, no. 11, 2022, doi: 10.3390/jimaging8110293.

[7] I. O. De Oliveira, R. Laroca, D. Menotti, K. V. O. Fonseca, and R. Minetto, "Vehicle-Rear: A New Dataset to Explore Feature Fusion for Vehicle Identification Using Convolutional Neural Networks," IEEE Access, vol. 9, pp. 101065–101077, 2021, doi: 10.1109/ACCESS.2021.3097964.

[8] A. Kherraki and R. El Ouazzani, "Deep convolutional neural networks architecture for an efficient emergency vehicle classification in real-time traffic monitoring," IAES Int. J. Artif. Intell., vol. 11, no. 1, pp. 110–120, 2022, doi: 10.11591/ijai.v11.i1.pp110-120.

[9] J. Zhu et al., "Vehicle Re-Identification Using Quadruple Directional Deep Learning Features," IEEE Trans. Intell. Transp. Syst., vol. 21, no. 1, pp. 410–420, 2020, doi: 10.1109/TITS.2019.2901312.

[10] R. Abiyev and M. Arslan, "Vehicle detection systems for intelligent driving using deep convolutional neural networks," Discov. Artif. Intell., vol. 3, no. 1, 2023, doi: 10.1007/s44163-023-00062-8.

[11] Y. Chen and Z. Li, "An Effective Approach of Vehicle Detection Using Deep Learning," Comput. Intell. Neurosci., vol. 2022, 2022, doi: 10.1155/2022/2019257.

[12] S. Ma and J. J. Yang, "Image-Based Vehicle Classification by Synergizing Features from Supervised and Self-Supervised Learning Paradigms," Eng, vol. 4, no. 1, pp. 444–456, 2023, doi: 10.3390/eng4010027.

[13] Y. Alotaibi, K. Nagappan, T. Thanarajan, and S. Rajendran, "Optimal deep learning based vehicle detection and classification using chaotic equilibrium optimization algorithm in remote sensing

imagery," Sci. Rep., vol. 15, no. 1, pp. 1–16, 2025, doi: 10.1038/s41598-025-02491-0.

[14] N. M. Alahdal, F. Abukhodair, L. H. Meftah, and A. Cherif, "Real-time Object Detection in Autonomous Vehicles with YOLO," Procedia Comput. Sci., vol. 246, no. C, pp. 2792–2801, 2024, doi: 10.1016/j.procs.2024.09.392.

[15] A. Chaudhuri, "Smart traffic management of vehicles using faster R-CNN based deep learning method," Sci. Rep., vol. 14, no. 1, pp. 1–11, 2024, doi: 10.1038/s41598-024-60596-4.

[16] Y. Djenouri, A. Belhadi, G. Srivastava, D. Djenouri, and J. Chun-Wei Lin, "Vehicle detection using improved region convolution neural network for accident prevention in smart roads," Pattern Recognit. Lett., vol. 158, pp. 42–47, 2022, doi: 10.1016/j.patrec.2022.04.012.

[17] S. Hayee, F. Hussain, M. H. Yousaf, M. Yasir, S. Ahmad, and S. Viriri, "A benchmark dataset and methodology for fine grained vehicle make and model classification," Front. Comput. Sci., vol. 7, 2025, doi: 10.3389/fcomp.2025.1561899.

[18] H. Zhang, X. Li, H. Yuan, H. Liang, Y. Wang, and S. Song, "A Multi-Angle Appearance-Based Approach for Vehicle Type and Brand Recognition Utilizing Faster Regional Convolution Neural Networks," Sensors, vol. 23, no. 23, 2023, doi: 10.3390/s23239569.

[19] U. Aulia, I. Hasanuddin, M. Dirhamsyah, and N. Nasaruddin, "A new CNN-BASED object detection system for autonomous mobile robots based on real-world vehicle datasets," Heliyon, vol. 10, no. 15, p. e35247, 2024, doi: 10.1016/j.heliyon.2024.e35247.

[20] P. K. Khoba, C. Parikh, C. V. Jawahar, R. K. Sarvadevabhatla, and R. Saluja, A Fine-Grained Vehicle Detection (FGVD) Dataset for Unconstrained Roadsg±, vol. 1, no. 1. Association for Computing Machinery, 2022. doi: 10.1145/3571600.3571626.

[21] Y. Hu, S. Li, J. Yan, W. Shao, and X. Luo, "Car-1000: A New Large Scale Fine-Grained Visual Categorization Dataset," 2025, [Online]. Available: http://arxiv.org/abs/2503.12385